Furthermore, even with users who are benign and careful, the mandatory security policy may still be subverted by flawed or malicious applications when only discretionary mechanisms are used to enforce it. The distinction between flawed and malicious software is not particularly important in this paper. In either case, an application may fail to apply security mechanisms required by the mandatory policy or may use security mechanisms in a way that is inconsistent with the user's intent. Mandatory security mechanisms may be used to ensure that security mechanisms are applied as required and can protect the user against inadvertent execution of untrustworthy applications.

Although the user may have carefully defined the discretionary policy to properly implement the mandatory policy, an application may change the discretionary policy without the user's approval or knowledge. In contrast, the mandatory policy may only be changed by the system security policy administrator.

In the case of personal computing systems, where the user may be the system security policy administrator, mandatory security mechanisms are still helpful in protecting against flawed or malicious software. In the simplest case, where there is only a distinction between the user's ordinary role and the user's role as system security policy administrator, the mandatory security mechanisms can protect the user against unintentional execution of untrustworthy software. With a further subdivision of the user's ordinary role into various roles based on function, mandatory security mechanisms can confine the damage that may be caused by flawed or malicious software.

Although there are a number of commercial operating systems with support for mandatory security, none of these systems have become mainstream. These systems have suffered from a fixed notion of mandatory security, thereby limiting their market appeal. Furthermore, these systems typically lack adequate support for constraining trusted applications. In order to reach a wider market, operating systems must support a more general